

China Business Trip Report Day 4

BB 087 VISUAL MACHINE

(PTC INTERNATIONAL)

Date : 16/05/2024

Meeting member Mr Siva (Maruwa)
 Mr Ronny (Maruwa Shanghai)
 Mr Miyano (Maruwa)
 Mr Kitauji (Maruwa)
 Mr Afiq (Maruwa)

CONFIDENTIAL

Discussion details

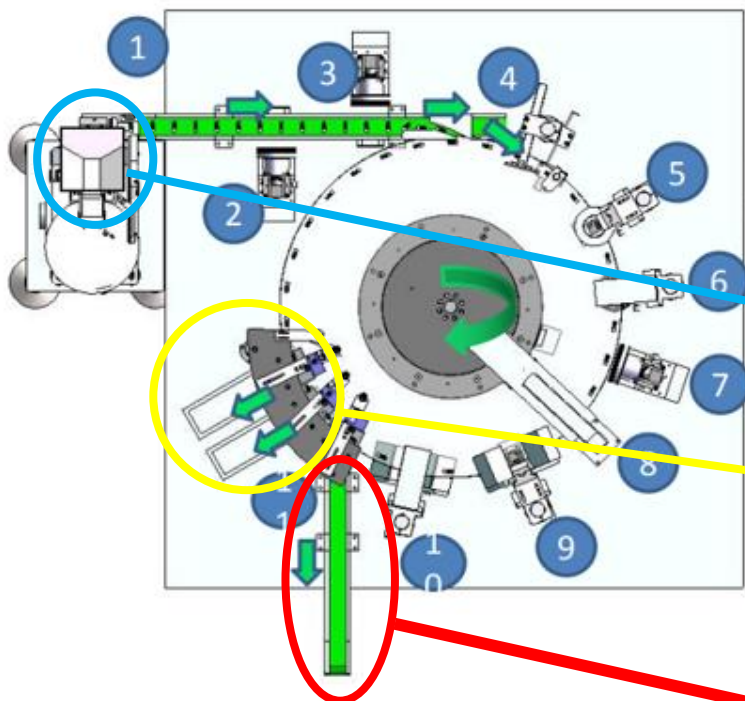
Date: 16/05/2024

Time : 9.00 am – 12.00 am



➔ Custom machine maker specialize
in Visual, Robotic, Inspection

➔ PTC CUSTOMER



BB 087 Visual Machine

PTC Visual Machine improvement

Product Type

BB-087

Machine Cycle

50,000 pcs / 24 hours

Additional Improvement

- Hopper size increase to 1 lot BB 087
- Additional sorting box (Uncritical NG, Critical NG, Undetermined)
- Output mechanism change (Product output stacking in height direction)

CONCLUSION BY TEAM



- ➔ Cuttermark NG limit reverified by Mr Kitauji
- ➔ Output mechanism change to BB 087 height stacking
- ➔ Requesting input hopper size to 8000 pcs – 9000 pcs BB 087
- ➔ For verification for missed NG, supplier requesting 1-5 lot product for verification testing
- ➔ Request to add additional output NG box from 2 to 3 (Critical NG, Uncritical NG, Undetermined)

China Business Trip Report Day 4

MO 464, MO 435 VISUAL MACHINE

(UNITECHNO)

Date : 16/05/2024

Meeting member Mr Siva (Maruwa)
 Mr Ronny (Maruwa Shanghai)
 Mr Miyano (Maruwa)
 Mr Kitauji (Maruwa)
 Mr Afiq (Maruwa)

CONFIDENTIAL

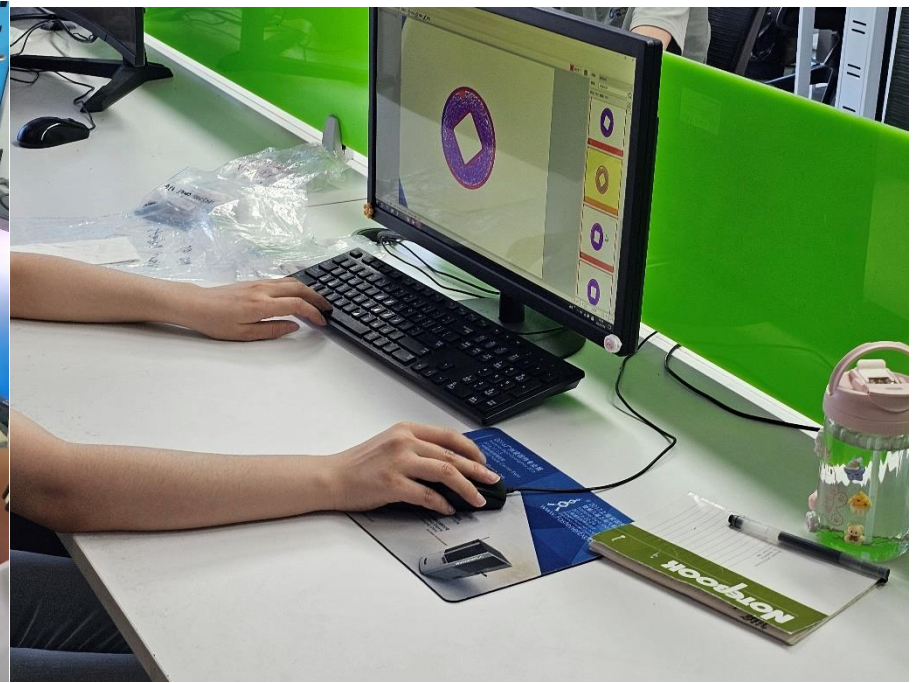
Discussion details

Date: 16/05/2024

Time : 3.30 pm – 6.30 pm



➔ Supplier specialize in AOI (Automated Inspection)
- standardize machine-



- ➔ Supplier have in house R&D team testing MO 464, MO 435 products
- ➔ Supplier have multiple lights, camera setup for products defects inspection



MEETING DISCUSSION BY SUPPLIER

- ➔ Camera 1 and 2 is for product dimension checking (ID/OD , Thickness)
- ➔ Camera 3 - 6 will have AI detection capabilities

MEETING DISCUSSION BY TEAM MEMBER

- ➔ Only use one type of machine for both MO 464, MO 457
(visual inspection without painting)
- ➔ MO 435, inner burr and katasuki NG unable to be check supplier will do testing with multiple light and camera angle.
- ➔ Cycle time will increase depending on number of light + camera



CONFIDENTIAL